fape-surgical-torch

PyTorch CUDA extension for the **FAPE Surgical Geometry** pass.

Install

```bash
pip install fape-surgical-torch/          # from repo root
# or after publishing to PyPI:
pip install fape-surgical-torch
```

Requires: Python ⩾ 3.8, PyTorch ⩾ 1.13, CUDA ⩾ 11.3, GPU sm_70+.

Usage

```python
import torch
from fape_surgical_torch import FapeSurgicalPass

passer = FapeSurgicalPass(tau=0.1, sigma_s=0.05, sigma_n=0.15, epsilon=1e-4)

vertices   = torch.randn(5000, 3, device="cuda", requires_grad=True)
faces      = torch.randint(0, 5000, (10000, 3), dtype=torch.int32, device="cuda")
all_cams   = torch.eye(4).unsqueeze(0).repeat(32, 1, 1).cuda()
hero_cams  = torch.eye(4).unsqueeze(0).repeat(4,  1, 1).cuda()

out_v, validity = passer(vertices, faces, all_cams, hero_cams)

loss = ((1.0 - validity) * out_v.norm(dim=-1)).mean()
loss.backward()
```

`vertex_normals` and `edges` are computed automatically if not supplied.

Tests

```bash
pytest tests/
```